

Project 4: Full Box–Jenkins study with unit-root testing

ADF/KPSS + differencing + SARIMA on real monthly data

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Context

Project 3 left us with a non-white-noise residual after a deterministic log-decomposition of `SouvenirSales.csv`. In this project you push the analysis to its conclusion: you classify the series as TS or DS using formal unit-root tests, apply the appropriate transformation (detrending or differencing), fit a SARIMA model, and produce a twelve-month forecast.

The deliverable is a complete, reproducible Box–Jenkins study that will serve as a template for Chapter 5 (ARMA) and Chapter 6 (volatility).

Deliverables

- Reproducible `project4_<name>.py` with seed 42.
- Report PDF (6–8 pages) including all figures, all numerical outputs, and a final forecast plot.
- Optional 8-min oral defence.

Exercise 1 – Visual diagnosis and transformation

- (1) Load `data/SouvenirSales.csv`. Plot the raw series Y_t . Comment on (i) the trend, (ii) the multiplicative seasonality (amplitude growing with the level).
- (2) Take logs: $y_t = \log Y_t$. Plot y_t . Verify visually that the seasonality is now *additive* (constant amplitude).
- (3) On y_t , plot the sample ACF up to lag 36. Identify the seasonal lags (12, 24, 36) and the slow decay typical of a non-stationary series.

Exercise 2 – Unit-root testing

- (1) Run the ADF test on y_t with three deterministic specifications:

- `regression="n"`: no constant, no trend;
- `regression="c"`: constant only;
- `regression="ct"`: constant and trend.

For each, report stat, p -value, 5% critical value, decision.

- (2) Run the KPSS test on y_t with `regression="c"` and `regression="ct"`. Report stat, p -value, decision.
- (3) Build the ADF/KPSS combination table. Which cell does y_t fall in? (TS, DS, or ambiguous?)

Exercise 3 – Differencing

- (1) Compute the regular difference $\Delta y_t = y_t - y_{t-1}$. Plot Δy_t and its ACF up to lag 36.
 (2) The seasonal lags should still dominate. Apply the seasonal difference $\Delta_{12} = (1 - L^{12})$ on top:

$$w_t = \Delta_{12} \Delta y_t = (1 - L)(1 - L^{12})y_t.$$

Plot w_t and its ACF/PACF up to lag 36.

- (3) Run ADF and KPSS on w_t (regression c). Confirm: w_t is stationary — y_t is SARIMA($\cdot, 1, \cdot$)($\cdot, 1, \cdot$) $_{12}$.

Exercise 4 – SARIMA model selection

- (1) From the ACF/PACF of w_t , propose two candidate SARIMA orders. The classical airline-style choice is $(0, 1, 1)(0, 1, 1)_{12}$ but you may also consider, e.g., $(1, 1, 0)(1, 1, 0)_{12}$ or $(1, 1, 1)(0, 1, 1)_{12}$.

- (2) Fit each candidate by maximum likelihood. Report:

- estimated coefficients (with standard errors);
- AIC and BIC;
- Ljung–Box p -value at lag 24 on residuals;
- residual standard deviation.

- (3) Pick the best model on the (AIC, BIC, Ljung–Box) triplet. Justify your choice in 5 lines.

Exercise 5 – Forecasting and back-transformation

- (1) On the chosen model, produce a 12-month forecast of $\log Y_t$ with 95% confidence intervals.
 (2) Back-transform: $\hat{Y}_{T+h} = \exp(\hat{y}_{T+h} + \hat{\sigma}^2/2)$ if you want the conditional mean of Y , or simply $\exp(\hat{y}_{T+h})$ for the conditional median. Plot the original Y_t together with the back-transformed forecast and CI on the original scale.
 (3) Discuss the interpretation: what does the model predict for sales in December of the year following 2001?

Exercise 6 – Bonus: hold-out validation

Split the data into a training set (1995–2000, 72 observations) and a test set (2001, 12 observations).

- Fit the chosen SARIMA on the training set.
- Compute the forecast errors on the test set.
- Report RMSE and MAPE on the original scale.

Discuss: does the model generalise out of sample?

Marking grid (out of 100)

Criterion	Points
Visual diagnosis + log transform	10
ADF and KPSS tests + combination matrix	20
Differencing $(1 - L)(1 - L^{12})$ + verification	15
SARIMA candidates + AIC/BIC/LB selection	25
Forecast + back-transform	15
Bonus: hold-out validation	10
Reproducibility + report quality	5
Total	100

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